

Hartshorne Chapter 1

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Section 1.2

Question 2.1

Consider $Z(\mathfrak{a})$ instead as a cone in $n + 1$ -affine space. Then as $\deg f > 0$ by assumption, $Z(f)$ cannot include the origin as it is homogeneous. Hence the behaviour of $Z(\mathfrak{a})$ at the origin can be ignored, and it is well-defined everywhere else. But then $f \in \sqrt{\mathfrak{a}}$ by the affine Nullstellensatz, and we are done. $\square$

Question 2.2

(i) $\Leftrightarrow$ (ii):

Consider $Z(\mathfrak{a})$ as a cone in affine $n + 1$ -space. Then $Z(\mathfrak{a}) = \emptyset$ in $\mathbf{P}^n$ iff $Z(\mathfrak{a}) = \{0\}$ or $\emptyset$ in $\mathbf{A}^{n+1}$. By the affine Nullstellensatz, $\sqrt{\mathfrak{a}}$ is going to be $\langle x_1, \dots, x_{n+1} \rangle$ or $k[x_1, \dots, x_{n+1}]$ respectively, which correspond to the homogeneous ideals S and $S_+ = \bigoplus_{d>0} S_d$ respectively.

(ii) $\Leftrightarrow$ (iii):

Forward direction is clear.

Now suppose that $S_d \subseteq \mathfrak{a}$ for some $d > 0$. Then $S_1 \subseteq \sqrt{S_d} \subseteq \sqrt{\mathfrak{a}}$. Hence $\mathfrak{a}$ is a homogeneous ideal that contains every positive degree homogeneous polynomial. Depending on if it contains a constant or not, $\sqrt{\mathfrak{a}} = S$ or S_+ . $\square$

Question 2.3

Skipped.

Question 2.4

Skipped.

Question 2.5

Skipped.

Question 2.6

WLOG suppose that we are working with x_0 and U_0 .

Let $\beta : k[x_1, \dots, x_n] \rightarrow S$ denote the homogenisation homomorphism as given in Proposition 2.2. Now consider some homogenised polynomial $\beta(f)$. Send $\beta(f)$ to $\frac{\beta(f)}{x_0}$, and let $\phi : k[x_1, \dots, x_n] \rightarrow S_{(x_0)}$ denote the composite map. It defines a ring homomorphism that sends $k[x_1, \dots, x_n]$ to the subring of $S_{(x_0)}$ of degree 0 elements.

Equally consider the evaluation ring homomorphism $\alpha : S \rightarrow k[x_1, \dots, x_n]$ given Proposition 2.2., and extend its domain to $S_{(x_0)}$.

To prove that ϕ and α define isomorphisms, it suffices to check monomials. Hence let $\frac{g}{x_0^k} \in S_{(x_0)}$ be such a monomial, such that $x_0 \nmid g$. Then $\phi(\alpha(g)) = \phi(g) = \frac{g}{x_0^{\deg g}}$ as desired. Equally if $f \in k[x_1, \dots, x_n]$ is a monomial we have that $\alpha(\phi(f)) = \alpha(\frac{f}{x_0^{\deg f}}) = f$ as desired.

Now note that if $\mathfrak{a} \triangleleft S$ is the radical ideal given by $I(Y)$, $A(Y_0)$ is exactly $k[x_1, \dots, x_n]/\alpha(\mathfrak{a})$. Hence by considering the isomorphism given above we get that $A(Y_p)$ is isomorphic to the subring of degree 0 elements in $S(Y)_{(x_0)}$. But then every fraction in $S_{(x_0)}$ can be multiplied by x_0 to the appropriate power, positive or negative, such that the result has degree 0. Hence $S(Y)_{(x_i)} \simeq A(Y_i)[x_i, x_i^{-1}]$ for all i .

Hence, by that result $Q(S(Y)) = Q(S(Y)_{(x_i)}) \simeq Q(A(Y_i)[x_i, x_i^{-1}]) = Q(A(Y_i))(x_i) \simeq Q(S(Y)_{(x_i)}|_{\deg=0})(x_i)$. But then by Theorem 1.8Aa, $\dim(S(Y)) = \text{trdeg}_k(Q(S(Y))) = \text{trdeg}(Q(S(Y)_{(x_i)}|_0)(x_i)) = \text{trdeg}(Q(S(Y)_{(x_i)}|_0)) + 1 = \dim(A(Y_i)) + 1 = \dim(Y_i) + 1$.

Moreover, as the U_i 's form a finite open cover for Y , we have that $\dim(S(Y)) = \dim(Y) + 1$. Finally notice that as the equality $\dim(Y) + 1 = \dim(S(Y)) = \dim(Y_i) + 1$ must hold for all i , as long as Y_i is nonempty such that $\dim(Y_i)$ is well-defined we get that $\dim(Y_i) = \dim(Y)$ necessarily. $\square$

Note. Sections of the proof are shamelessly copied from <https://williamtroiani.github.io/pdfs/HartshorneSolutions.pdf> because there was no way I was figuring out the dimensionality argument.

Question 2.7

Skipped.

Question 2.8

Notice that a projective variety $Y \subseteq \mathbf{P}^n$ has dimension $n - 1$ iff $\dim(S(Y)) = n$ by the result of 2.6 above. But then $\dim(S(Y)) = n$ iff the prime ideal $\mathfrak{p} := I(Y)$ necessarily has height one in $k[x_1, \dots, x_n]$ by Theorem 1.8A. But then by Prop 1.12A this is equivalent to $\mathfrak{p} = \langle f \rangle$ for some irreducible f . Hence Y is a hypersurface iff it has dimension $n - 1$. $\square$

Question 2.9

a):

That $\beta(I(Y)) \subseteq I(\bar{Y})$ is clear.

Now suppose that $f \in I(\bar{Y})$ is a homogeneous polynomial that vanishes on Y . Then $\alpha(f) \in I(Y)$, but $\beta(\alpha(f)) = f$ ($\beta \circ \alpha$ is not the identity in general but it is on homogeneous elements). Hence we get equality on the homogeneous terms, such that $I(\bar{Y}) = \langle \beta(I(Y)) \rangle$ as it is necessarily generated by homogeneous elements. $\square$

b):

From Exercise 1.2 we have that the twisted cubic has ideal $\langle y - x^2, z - x^3 \rangle$. But the projective closure of the twisted cubic has ideal generated by $\langle wy - x^2, wz - x^3, xy - zw \rangle$. $\square$

Note. Lazy proof I know, but trust me when I say I did spent time thinking through this problem, and I think I get the important ideas. See the dm I sent to you for my main takeaways.

Question 2.10

Skipped.

Question 2.11

a)

Let $\{p_i\}$ denote a family of pairwise coprime linear polynomials.

Then $Y = \bigcap_i Z(p_i) \iff Y = Z(\bigcup_i p_i) \iff I(Y) = \langle p_1, \dots, p_k \rangle$. $\square$

b)

Suppose that $Y \subseteq \mathbf{A}^n$ is a linear variety of dimension r . Then we have that $\text{ht}(I(Y)) = n - r$. But then if $I(Y)$ is generated by $m < n - r$ linear polynomials, by Krull's height theorem we have that $\text{ht}(I(Y)) \leq m$ which is a contradiction.

Now note that the zero set of a linear polynomial describes an affine subspace of $\mathbf{A}^n$. And as intersections of affine planes is still an affine plane, we get a correspondence between linear varieties and affine spaces.

Hence, by considering affine subvarieties of Y , we get that the dimension of Y , considered as an affine vector space over k , is less than or equal to the dimension of Y as a topological space. Thus if we have a collection of linear polynomials that vanish on Y that generated $I(Y)$, at most $n - r$ of them can be k -linearly independent. Hence $I(Y)$ is minimally generated by $n - r$ linear polynomials. $\square$

c)

By (b) above we have that $I(Y)$ and $I(Z)$ are generated by $n - r$ and $n - s$ polynomials respectively. Hence $I(Y \cap Z)$ is generated by $2n - r - s$ linear polynomials, such that $Y \cap Z$ has dimension at least $n - (2n - r - s) = r + s - n > 0$ by Question 1.9. Hence it cannot be empty. Furthermore by the reasoning given in (b), $Y \cap Z$ is a linear variety, and we are done. $\square$

Question 2.12

a)

As the image is embedded within an integral domain, that the kernel is a prime ideal is clear.

Suppose that $f \in \ker \theta$, and let m_i denote the sum of all homogeneous terms of degree i in f , such that we get the decomposition $f = \sum_i m_i$. Then we get that $0 = \theta(f) = \sum_i \theta(m_i)$; and as the degree of two monomials under θ are the same iff they have the same degree we have that $\theta(m_i) = 0$ for all i . Hence $\ker \theta$ is a prime homogeneous ideal, so $Z(\ker \theta)$ defines a projective variety in $\mathbf{P}^N$. $\square$

b)

That $\rho_d(\mathbf{P}^n) \subseteq Z(\ker \theta)$ is clear: Suppose that $\vec{p} = (M_0(\vec{a}), \dots, M_N(\vec{a})) \in \rho_d(\mathbf{P}^n)$ for some $\vec{a} \in \mathbf{P}^n$. Then for any $f \in \ker \theta$ such that $\theta(f) = f(M_0, \dots, M_N) = 0$, we automatically have that $f(\vec{p}) = f(M_0(\vec{a}), \dots, M_N(\vec{a})) = 0$.

Now, notice that if $x_0^{n_0} \dots x_n^{n_n}$ denotes some arbitrary monomial with degree d , we have that $(x_0^{n_0} \dots x_n^{n_n})^d = (x_0^d)^{n_0} \dots (x_n^d)^{n_n}$. Now let $\vec{p} \in Z(\ker \theta) \setminus \rho_d(\mathbf{P}^n)$. Consider the entries in $\vec{p}$ with index corresponding to the monomials $x_0^d, x_1^d, \dots, x_n^d$, and label them $p_0, \dots, p_n$. Then using the entries p_i as a fixed reference, there must exist a coordinate p' of $\vec{p}$, corresponding to the monomial $x_0^{n_0} \dots x_n^{n_n}$, such that $p'^d \neq (p_0^d)^{n_0} \dots (p_n^d)^{n_n}$. But then $\vec{p} \notin Z((x_0^{n_0} \dots x_n^{n_n})^d - (x_0^d)^{n_0} \dots (x_n^d)^{n_n})$, which is a contradiction. Hence $\rho_d(\mathbf{P}^n) \supseteq Z(\ker \theta)$, and we get equality. $\square$

c)

That ρ_d is bijective is clear. That it is a continuous map is also clear, as if $f \in k[y_0, \dots, y_N]$ then $\rho_d^{-1}(Z(f))$ is exactly $Z(\theta(f))$.

Now suppose $f \in k[x_0, \dots, x_n]$ is a homogeneous polynomial. We construct an ideal $\mathfrak{a}$ in $k[y_0, \dots, y_N]$ such that $\rho_d(Z(f)) = Z(\mathfrak{a})$. This is sufficient, as every ideal is finitely generated, such that for an arbitrary closed set Y we have that $I(Y) = \langle f_1, \dots, f_m \rangle = \langle \bigcup_i f_i \rangle = \bigcap_i Z(f_i)$.

Pick a $k \in \mathbb{Z}$ such that $d \mid k + \deg f$. Then for every i , $f_i := x_i^k f$ is a polynomial in the degree d monomials and is hence contained in the image of θ . So let $\mathfrak{a} := \theta^{-1}(\langle f_i \rangle_{i=0}^n)$. Then if $\rho_d(\vec{p}) \in Z(\mathfrak{a})$, WLOG let p_0 be a nonzero coordinate of $\vec{p}$. Then $f_0(\vec{p}) = p_0^k f(\vec{p}) = 0$ and thus $f(\vec{p}) = 0$. Hence $\rho_d(Z(\mathfrak{a})) = Z(f)$ as well. Thus ρ_d is a closed map; and as it is a continuous bijection it is also a homeomorphism. $\square$

d)

Label the 3-monomials in two variables as $M_0 := x^3, M_1 := x^2y, M_2 := xy^2, M_3 := y^3$. Then the 3-uple embedding of $\mathbf{P}^1$ in $\mathbf{P}^3$ is exactly the set $(M_0 : M_1 : M_2 : M_3)$. As ρ_d defines a homeomorphism this image is irreducible; and by (b) above image is closed. But then it contains the twisted cubic such that both have dimension 1 so they are the same. $\square$

Question 2.13

As Z is a curve in Y , $\rho_2^{-1}(Z)$ is a curve in $\mathbf{P}^2$ via the homeomorphism constructed in 2.12. Hence write $\rho_2^{-1}(Z) = Z(f)$ for some homogeneous irreducible $f \in k[x, y, z]$. If f has even degree then $f = \theta(g)$ for some $g \in k[x_0, \dots, x_5]$. Otherwise f^2 is an even degree homogeneous polynomial with $Z(f^2) = Z(f)$, such that there is a g where $\theta(g) = f^2$. That g is irreducible in the first case is clear. In the second case, note that any decomposition of g corresponds to a decomposition of f^2 ; but the only one available is $f^2 = f \cdot f$, where f cannot be in the image of θ . Hence g is always irreducible.

Thus, the bijection proven in 2.12 above gives us that $Z = Y \cap Z(g)$, where $Z(g)$ is the hypersurface as desired.

Question 2.14

Let the homogeneous coordinates of $\mathbf{P}^N$ be denoted by (z_{ij}) , and consider the homomorphism $\theta : k[z_{ij}] \rightarrow k[x_0, \dots, x_r, y_0, \dots, y_s]$ given by $z_{ij} \mapsto x_i y_j$. Note that $\ker \theta$ is clearly a prime ideal as the image is contained in an integral domain. Next we show that $Z(\ker \theta) = \text{Im}(\psi)$. The proof proceeds as very similar to that in 2.12(b):

That $\text{Im}(\psi) \subseteq Z(\ker \theta)$ is clear: Suppose that $\vec{p} = (p_{x_i} p_{y_j}) \in \text{Im}(\psi)$. Then for any $f \in \ker \theta$ such that $\theta(f) = f(x_i y_j) = 0$, we automatically have that $f(\vec{p}) = f(p_{x_i} p_{y_j}) = 0$.

Suppose that $\vec{p} \in Z(\ker \theta) \setminus \text{Im}(\psi)$. Then, using the entries p_{0j} and p_{i0} of $\vec{p}$ as fixed reference, there exists some p_{kl} such that $p_{00} p_{kl} \neq p_{0l} p_{k0}$. But then $\vec{p} \notin Z(z_{00} z_{kl} - z_{0l} z_{k0})$, where $\theta(z_{00} z_{kl} - z_{0l} z_{k0}) = x_0 y_0 x_k y_l - x_0 y_l x_k y_0 = 0$; which is a contradiction. Hence $\text{Im}(\psi) \supseteq Z(\ker \theta)$, and we get equality. $\square$

Question 2.15

a)

Define the embedding $\phi : \mathbf{P}^1 \times \mathbf{P}^1 \hookrightarrow \mathbf{P}^3$ where $(x_0 : x_1), (y_0 : y_1) \mapsto (x_0 y_0 : x_0 y_1 : x_1 y_0 : x_1 y_1)$. Then $x_0 y_1 x_1 y_0 - x_0 y_0 x_1 y_1 = 0$, and a similar proof as done in 2.14 yields that $\text{Im}(\phi) = Z(xy - zw)$. $\square$

b)

In both copies of $\mathbf{P}^1$, notice that we can WLOG assume the first entry is nonzero such that every point in the space has the form $(1 : t)$ where t ranges over every value in k .

If t is fixed for the first component, then its image under the Segre embedding corresponds to the set $(y_0 : y_1 : t y_0 : t y_1)$ where $(y_0 : y_1)$ ranges over $\mathbf{P}^1$. This is a linear projective variety, with ideal given by $\langle y - tw, z - tx \rangle$. It can be seen that the ideal has height 2, so the projective variety has dimension 1 as desired. The construction for the second component is similar, and these give the families $\{L_t\}$ and $\{M_t\}$.

Now if $u \neq t$, notice that the projective variety $L_t \cap L_u$ has ideal given by $\langle y - tw, z - tx, y - uw, z - ux \rangle$. But then these are 4 algebraically independent linear polynomials in $k[w, x, y, z]$ such that they generate the entire ring. Hence by the homogeneous Nullstellensatz the intersection is empty. The proof for $M_t \cap M_y = \emptyset$ is analogous.

Now consider $L_u \cap M_t$. The ideal of the intersection is given by $\langle y - tw, z - tx, w - ux, z - uy \rangle$. If both $t, u = 0$ then it is clear that the coordinate ring is isomorphic to k . Otherwise suppose that $t \neq 0$, such that in the coordinate ring we have that $w \equiv ux \equiv \frac{u}{t}z \equiv \frac{u^2}{t}y \equiv u^2w$, such that $w \equiv 0$. But then this proof applies for x, y, z as well such that the coordinate ring is also isomorphic to k . Hence the ideal is a maximal ideal, and by the homogeneous Nullstellensatz is exactly given by the kernel of the evaluation homomorphism at some point $\vec{p} \in \mathbf{P}^3$. In other words, $L_u \cap M_t = Z(\langle y - tw, z - tx, w - ux, z - uy \rangle) = \{\vec{p}\}$. $\square$

Question 2.16

a)

Let $Q_1 := Z(x^2 - yw)$ and $Q_2 := (xy - zw)$, both considered as projective varieties in $\mathbf{P}^3$.

Then the points in the intersection $Q_1 \cap Q_2$ are given by $(w : x : \frac{x^2}{w} : \frac{x^3}{w^2}) = (w^3 : w^2x : wx^2 : x^3)$ if $w \neq 0$, which corresponds to a twisted cubic, and $(0 : 0 : y : z)$ if $w = 0$, which is a line. $\square$

b)

Let $C := Z(x^2 - yz)$ and $L := Z(y)$, both considered as projective varieties in $\mathbf{P}^3$.

Then the points in the intersection are given by $(0 : 0 : z)$, i.e. is a single point. But then the sum of the ideals is $\langle x^2, y \rangle$, which is not a prime ideal and hence not the minimal ideal corresponding to $C \cap L$. $\square$

Question 2.17

a)

Suppose that $Y \subseteq \mathbf{P}^n$ is a projective ideal such that $\mathfrak{a} := I(Y)$ can be generated by q elements. Then Krull's height theorem gives us that $\text{ht}(\mathfrak{a}) \leq q$, such that by the result of 2.6 it immediately follows that $\dim(Y) \geq n - q$. $\square$

b)

Suppose that Y is a strict complete intersection such that $I(Y) = \langle f_1, \dots, f_q \rangle$. Then $Y \supseteq \bigcap_i Z(f_i) \supseteq Z(\langle f_1, \dots, f_q \rangle) = Y$, such that Y is also a set-theoretic complete intersection. $\square$

c)

Skipped.

d)

Nope.